

CORE 200 Scientific Methods

Handout 12

Group name

Enter your details :

Section

Does it contain any abstract reasoning concerning quantity or number? No. Does it contain any experimental reasoning concerning matter of fact and existence? No. Commit it then to the flames: for it can contain nothing but sophistry and illusion.

David Hume, *A Treatise of Human Nature*, 1739.

Assume that whatever you want to study has, not causes, but a history, a story, a narrative, a 'first this happened, then that happened, and then the other happened, and it ended up like this.' On this view we understand the occurrence of events by learning the steps in the process by which they came to happen, rather than by learning the conditions that made their existence necessary.

Howard Becker (sociologist), *Tricks of the Trade*, 1998.

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.



Reflect and discuss (A)



1.



2.



3.



4.



5.



6.

 7.

 **Reflect and discuss (B)**

 8.

 9.

 10.

 11.

 12.

 **Reflect and discuss (C)**

See the algorithm in Fig. 2 of Barroga and Matanguihan (2022) for building a research question/hypothesis in quantitative research. Carry out steps 1 and 2 with the assistance of your instructor.

Even though these are preliminary steps, you will also need to think ahead a bit. For that purpose, also see the helpful list of questions on page 3.

TABLE 2.3 *Organizing a quantitative research study*

PROBLEM

What is the goal of the research?
What is the problem, issue, or critical focus to be researched?
What are the important terms? What do they mean?
What is the significance of the problem?
Do you want to test a theory?
Do you want to extend a theory?
Do you want to test competing theories?
Do you want to test a method?
Do you want to replicate a previous study?
Do you want to correct previous research that was conducted in an inadequate manner?
Do you want to resolve inconsistent results from earlier studies?
Do you want to solve a practical problem?
Do you want to add to the body of knowledge in another manner?

REVIEW OF LITERATURE

What does previous research reveal about the problem?
What is the theoretical framework for the investigation?
Are there complementary or competing theoretical frameworks?
What are the hypotheses and research questions that have emerged from the literature review?

SAMPLE

Who (what) will provide (constitute) the data for the research?
What is the population being studied?
Who will be the participants for the research?
What sampling technique will be used?
What materials and information are necessary to conduct the research?
How will they be obtained?
What special problems can be anticipated in acquiring needed materials and information?
What are the limitations in the availability and reporting of materials and information?

METHOD

What methods or techniques will be used to collect the data? (This holds for applied and non-applied research)
What procedures will be used to apply the methods or techniques?
What are the limitations of these methods?
What factors will affect the study's internal and external validity?
Will any ethical principles be jeopardized?

DATA ANALYSIS

How will data be analysed?
What statistics will be used?
What criteria will be used to determine whether hypotheses are supported?
What was discovered (about the goal, data, method, and data analysis) as a result of doing preliminary work (if conducted)?

CONCLUDING INFORMATION

How will the final research report be organized? (Outline)
What sources have you examined thus far that pertain to your study? (Reference list)
What additional information does the reader need?
What time frame (deadlines) have you established for collecting, analysing and presenting data? (Timetable)

Adapted from Rubin et al. (1990)